Ogbonna Arinze

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SUMMARY

Pragmatic, product-minded Senior Fullstack Engineer with 8+ years of experience building high-impact web applications using TypeScript, React, Next.js, Tailwind, and Node.js, with backend expertise in Drizzle ORM, PostgreSQL, and Vercel deployments. Passionate about designing production-grade LLM-powered systems, including agentic workflows, complex prompt chains, and knowledge-capture tools using the Vercel AI SDK and LangChain. Thrives in early-stage, high-autonomy environments, combining strong UX intuition with deep technical ability to own product features end-to-end, from architecture decisions through to user-facing delivery.

CORE COMPETENCIES

- **Fullstack Development:** TypeScript, React, Next.js, Tailwind CSS, Node.js, Drizzle ORM, PostgreSQL (Neon/Serverless Postgres), and Vercel.
- **AI & LLM Systems:** Agentic workflows, prompt engineering, complex prompt chains, Vercel AI SDK, LangChain, Mastra, RAG strategies, embeddings and vector search, OpenAI API, and Anthropic API.
- **Product & UX Engineering:** Product-minded feature ownership, UX-driven design decisions, accessibility (WCAG), design systems, and Figma-to-code workflows.
- **Frontend Architecture:** SSR/SSG/ISR strategies, component libraries, micro-frontends, monorepos (Turborepo, Nx), and state management (Redux, Zustand, React Query).
- **Performance & Observability:** Core Web Vitals optimization, code splitting, lazy loading, CDN strategies, Sentry, Datadog RUM, and Lighthouse profiling.
- **CI/CD & Quality:** GitHub Actions, automated testing (Jest, Cypress, Playwright), Docker, feature flags, and canary deployments.
- **API & Data Layer:** RESTful APIs, GraphQL, real-time APIs (WebSockets), JWT authentication, role-based access control, and schema design.
- **Leadership & Methodology:** High-autonomy ownership, technical architecture reviews, mentorship, Agile/Scrum, and TDD/BDD practices.

WORK EXPERIENCE

Roche Genentech — San Francisco, CA (Remote)

Senior Fullstack Engineer (React / Next.js / Node.js) — *May 2023 - Present*

- Architected the **fullstack platform** for a clinical trial data application using **React, Next.js, TypeScript**, and **Tailwind CSS** on the frontend, with **Node.js** API services and **PostgreSQL** on the backend, deployed and monitored on **Vercel**.
- Designed and built a **reusable design system** in **React** and **Radix UI** with **Tailwind CSS** styling, improving UI consistency, accessibility (WCAG), and developer velocity across multiple product teams.
- Developed **LLM-powered knowledge capture tools** using the **Vercel AI SDK** and **LangChain**, enabling domain experts to structure clinical trial workflows through **agentic prompt chains** that automated data classification and enrichment.
- Implemented **server-side data processing pipelines** in **Node.js** and **TypeScript**, handling large-scale clinical datasets with validation layers and automated enrichment workflows to ensure data quality and integrity.
- Enhanced application security and UX by engineering **JWT-based authentication** and **role-based access control** in a **Next.js** fullstack application with **React Query**, using HTTP-only cookies for secure token handling and enforcing permission-based rendering.
- Optimized application performance by implementing **lazy loading, code splitting, dynamic imports**, and **CDN asset delivery**, reducing load times significantly and improving **Core Web Vitals** scores measured via **Lighthouse** and Chrome DevTools.
- Scaled **state management** in a large **Next.js** application by integrating **Redux** for complex client state and **React Query** for cached API data, eliminating redundant re-renders and establishing maintainable, well-documented patterns.
- Transformed a **legacy Vue.js web application** into a **React Chrome extension**, collaborating directly with designers and backend engineers to optimize clinical site workflows and drive adoption through superior UX.

- Increased release reliability by integrating **Playwright end-to-end tests** into the **GitHub Actions CI/CD pipeline**, automating test reporting with **Allure Reports** and configuring Slack notifications for real-time build quality visibility.
- Owned product features end-to-end with **high autonomy**, from initial architecture decisions and database schema design through frontend implementation and production deployment, working closely with product stakeholders to shape the roadmap.

DESCASIO LTD Lagos, Nigeria
Staff Fullstack Engineer *May 2021 - Apr 2022*

- Owned the **fullstack architecture** for a data-heavy **React** dashboard with **Node.js** backend services and **PostgreSQL**, implementing virtualization and efficient state management to handle large-scale datasets with real-time updates.
- Designed and implemented **RESTful APIs** and database schemas to support complex data workflows, optimizing query performance and ensuring seamless data flow between the frontend and backend layers.
- Optimized **fullstack performance** by profiling render bottlenecks with **React Profiler** and **Lighthouse Audits**, reducing bundle size by **30%-40%** through rollup-plugin-visualizer and webpack-bundle-analyzer, achieving a **35% faster load time**.
- Built **product features autonomously** in an early-stage, fast-paced environment, making independent architecture decisions and iterating rapidly based on user feedback to deliver high-impact functionality.
- Established engineering standards for **code quality, testing**, and deployment across the team using **ESLint, Prettier, Husky, Lint-staged**, and **GitHub Actions**, reducing production errors and accelerating CI/CD processes.
- Championed **UX-driven development** practices, collaborating with designers to ensure every feature met high usability standards, conducting design reviews, and advocating for intuitive user interactions across the product.
- Mentored junior engineers on **fullstack best practices**, clean code principles, and design patterns, fostering a culture of high-quality, maintainable code and consistent design across projects.

EDUCATION

University of Nigeria Nsukka, Enugu
BSc Computer Science - 4.1 CGPA *Graduation Date: Sep 2019*